

FP1 Chapter 7 Matrices (1)

Lecture 7 Matrices (1)

(Matrix Algebra, Singular, Determinant, Inverse, Transformation, Invariant)

(A) MATRIX ALGEBRA

1. Introduction:

- In a class there are three friends Anson, Benjamin, Casey. They have 15, 10 and 13 notebooks and 6, 2, 5 pens respectively. We can represent this information in a rectangular array as follows:

- A is called a matrix of order 3×2 (3 by 2), 3 rows and 2 columns.
- The same information can be also represented as:

- B is called a matrix of order 2×3 (2 by 3), 2 rows and 3 columns.



2. General Form of a Matrix:

- A matrix of order $m \times n$:

3. Square Matrix: A matrix having the same number of rows and columns.

4. Diagonal Matrix: A square matrix in which all non-diagonal elements are zero.

5. Row matrix: Only one row.



6. **Column matrix: Only one column.**
7. **Zero matrix: A matrix whose all the elements are zero.**
8. **Identity matrix: A square matrix in which all the diagonal element are 1 each and the rest are 0.**
9. **Equal matrix: Two matrices are equal if they have the same order and the corresponding elements are equal.**



10. Scalar multiplication:

11. Addition and Subtraction of Matrices:

- (i) Both the matrices should be of the same order.
- (ii) Add (Subtract) the corresponding elements.

- Properties of addition of matrices:

Given matrices of the same order: (i) Commutative:

(ii) Associative:

12. Multiplication of Two Matrices:

Condition: $A_{m \times n} \times B_{n \times p} = C_{m \times p}$

Two matrices A and B can only be multiplied if the number of columns in the first matrix is the same as the number of rows in the second matrix.

- Matrix multiplication is not commutative in general:

$$A_{2 \times 2} \times B_{2 \times 2} = C_{2 \times 2}$$

$$B_{2 \times 2} \times A_{2 \times 2} = D_{2 \times 2}$$

- $AB \neq BA$ in general.
- But matrix multiplication is associative: $(AB)C = A(BC)$.

13. Determinant of order 2:

- To each squared matrix there corresponds a value, determinant of matrix A is denoted by $\det A$ or $|A|$.
- Given a squared matrix of order 2, $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $\det A$ or $|A| = ad - bc$.

14. Determinant of Order 3:

- Given a squared matrix A of order 3×3 ,

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

- We evaluate the value of $\det A$, by expanding with first row / first column.

$$\det A = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

- **Example 1: Find the determinant of the matrix.**

$$(i) \quad A = \begin{pmatrix} -1 & 6 & -2 \\ 2 & 1 & 1 \\ 4 & 1 & -3 \end{pmatrix}$$

$$(ii) \quad M = \begin{pmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 2 & 5 & 2 \end{pmatrix}$$

15. Singular Matrix:

- A squared matrix A is said to be **singular** if $\det A = 0$. (If $\det A \neq 0$ then A is non-singular matrix.)
- **Example 2: Determine if the matrix is singular.**

$$(i) \quad A = \begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix}$$

$$(ii) \quad M = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 3 & 2 \\ 3 & 2 & 3 \end{pmatrix}$$

16. Scalar multiplication:

Given a matrix A and real number k , then kA is obtained by multiplying each element of A by k .

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17. Inverse of a matrix of order 2:

Given a square matrix A of second order $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then another square matrix B of the same order, such that:

- **Example 3:**

Given $M = \begin{pmatrix} 3 & 1 \\ -11 & -2 \end{pmatrix}$, find M^{-1} .

- Note: If A and B are two square matrices of the same order, $|A| \neq 0, |B| \neq 0$.
Then $(AB)^{-1} = B^{-1}A^{-1}$.

- **Example 4:**

Given two invertible matrices A and B , find A^{-1} , B^{-1} , and hence find $(AB)^{-1}$.

$$A = \begin{pmatrix} 3 & 2 \\ 7 & 5 \end{pmatrix}, B = \begin{pmatrix} 4 & 6 \\ 3 & 2 \end{pmatrix}.$$

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18. Transpose of a Matrix:

- Given a matrix $A_{m \times n}$ then its transpose denoted by A^T (or A') is obtained by interchanging its rows and columns, and its order will be $n \times m$.

- **Example 5:**

Given $A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 4 \\ 2 & 5 \end{pmatrix}$, verify $(AB)^T = B^T A^T$.

19. Minor and Cofactor of the element of a Matrix:

Given $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$

- Minor of element is the value of the determinant obtained by deleting the element of row and column.

- Cofactor of an element is the value of corresponding minor with proper sign.

- **Example 6:**

Write the values of minors and cofactors of each element of the first column.

$$A = \begin{pmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{pmatrix}$$

20. Adjoint of a Matrix of Order 2:

- Given a square matrix A , $\text{adj } A$ is the transpose of a matrix, whose elements are cofactors of the respective elements of given matrix A .

- **Example 7:**

Given a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Find $\text{adj } A$.

- Now consider,

21. Adjoint of a Matrix of Order 3:

$$\text{Given } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

- Example 8:

$$\text{Given a matrix } A = \begin{pmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{pmatrix}, \text{ find}$$

- (i) $\text{adj } A$ (ii) A^{-1} (iii) Verify that $AA^{-1} = I$. (iv) $|\text{adj } A| = |A|^2$.



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- $(adj AB) = (adj A)(adj B)$

- $|AB| = |A||B|$

- $|A^{-1}| = \frac{1}{|A|}$

- **Example 9:**

A, B, C are matrices of order 2.

(i) Given $BAC = I$, show $A = B^{-1}C^{-1}$.

(ii) If $AX = B$, show $X = A^{-1}B$

(iii) Given a squared matrix A , which satisfy $x^2 - 6x + 17 = 0$, find A^{-1} .

(iv) Given $M = \begin{pmatrix} 3 & 1 \\ -11 & -2 \end{pmatrix}$, verify that $|M^{-1}| = \frac{1}{|M|}$.

22. Some properties of Determinants:**- Example 10:**

Without expanding find the value of determinant.

$$(i) \quad D = \begin{pmatrix} 4 & 8 & 12 \\ 1 & 2 & 3 \\ 7 & -5 & 1 \end{pmatrix}$$

$$(ii) \quad M = \begin{pmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{pmatrix}$$

**(B) GEOMETRIC TRANSFORMATIONS IN TWO DIMENSIONS**

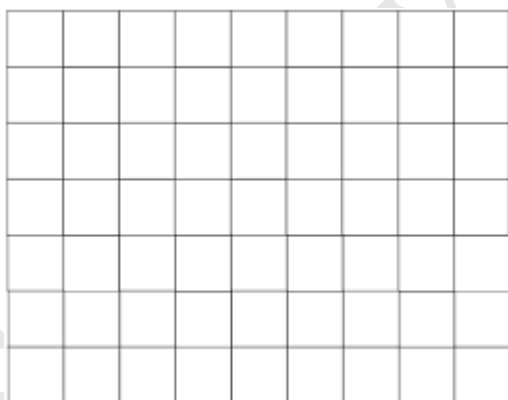
- With the application of a transformation, a shape changes its **position** or **size** or **both**. The original shape is called **object** and the new shape is called its **image**.

1. Reflection transformation:

- Reflection is the flip of shape over the line of reflection, called **mirror line, M** . Any point of the object and the corresponding point on the image are equidistant from the mirror line, along a line perpendicular to the mirror line.

(I) Reflection in the x -axis

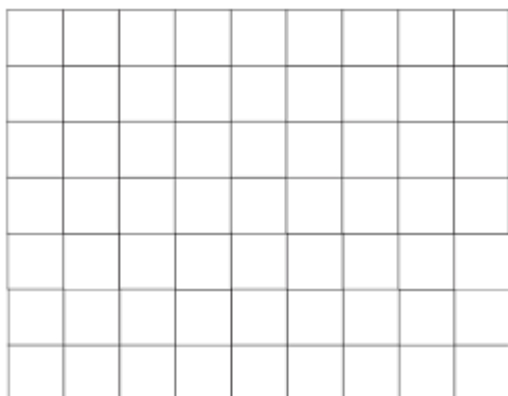
- x -axis is the mirror line M .





(II) Reflection in the y -axis

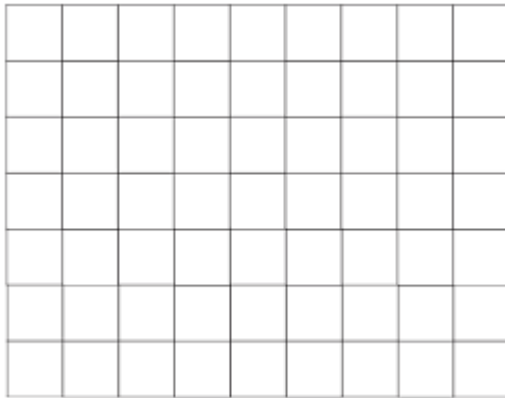
- y -axis is the mirror line M .



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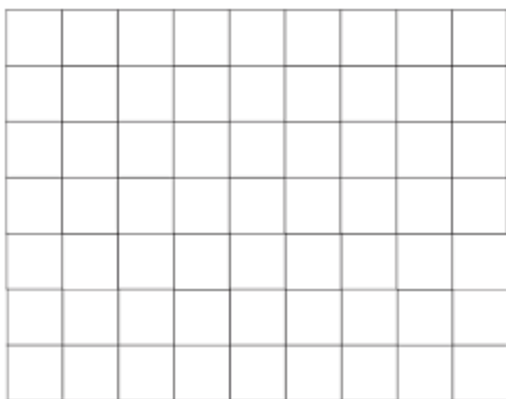
(III) Reflection in the origin



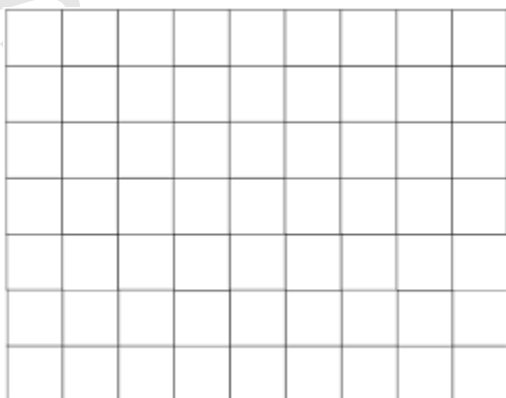
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(IV) Reflection in the line $y = x$



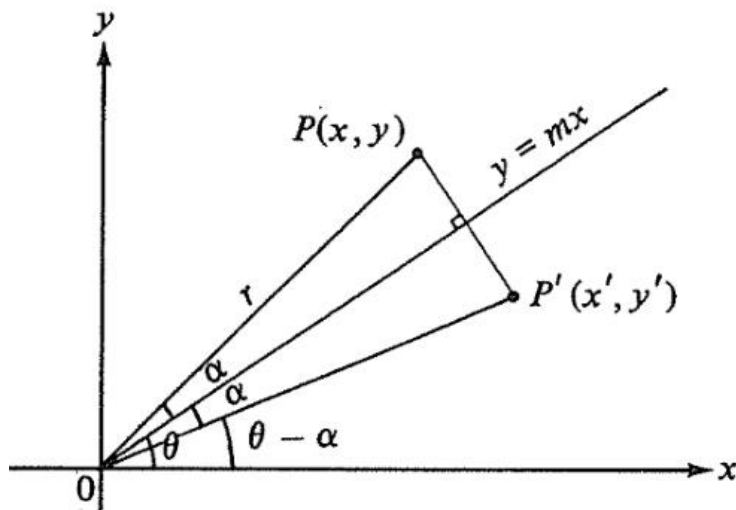
(V) Reflection in the line $y = -x$





(VI) Reflection in the line $y = (\tan \theta)x$

- Take a point $P(x, y)$ in the plane. Let $P(x', y')$ be the image of P under the reflection in the line $y = (\tan \theta)x$.



- For point P : $x = r \cos(\alpha + \theta)$, $y = r \sin(\alpha + \theta)$. For point P' : $x' = r \cos(\theta - \alpha)$, $y' = r \sin(\theta - \alpha)$.
- But then, $x' = r \cos(\theta - \alpha)$

$$= r \cos[(\alpha + \theta) - 2\theta]$$

$$= r[\cos(\alpha + \theta) \cos 2\theta + \sin(\alpha + \theta) \sin 2\theta]$$

$$= x \cos 2\theta + y \sin 2\theta$$
- Also, $y' = r \sin(\theta - \alpha)$

$$= -r \sin[(\alpha + \theta) - 2\theta]$$

$$= -r [\sin(\alpha + \theta) \cos 2\theta - \cos(\alpha + \theta) \sin(2\theta)]$$

$$= -y \cos 2\theta + x \sin 2\theta$$

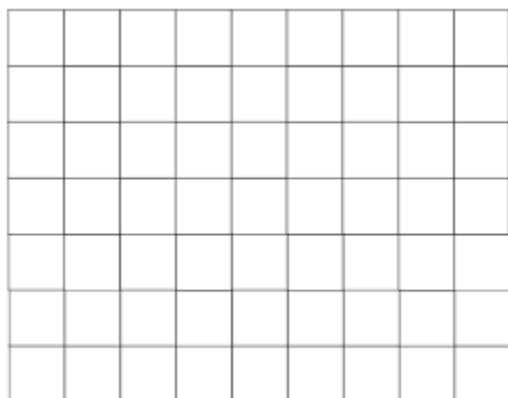
$$= x \sin 2\theta - y \cos 2\theta$$
- It follows that $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$. Hence matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$ represents the reflection in the line $y = (\tan \theta)x$.

- For reflection in the x -axis, $\theta = 0^\circ$, the required matrix is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.
- For reflection in the y -axis, $\theta = 90^\circ$, the required matrix is $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- For reflection in the line $y = x$, $\theta = 45^\circ$, the required matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- For reflection in the line $y = \sqrt{3}x$, $\theta = 60^\circ$, the required matrix is $\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$.
- For reflection in the line $y = mx$, the required matrix is $\begin{pmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & -\frac{1-m^2}{1+m^2} \end{pmatrix}$.



2. Rotation Transformation

(i) Rotation through an angle θ anticlockwise about the origin



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(II) Rotation through an angle θ clockwise about the origin

(III) Rotation through an angle 180° clockwise/anticlockwise about the origin

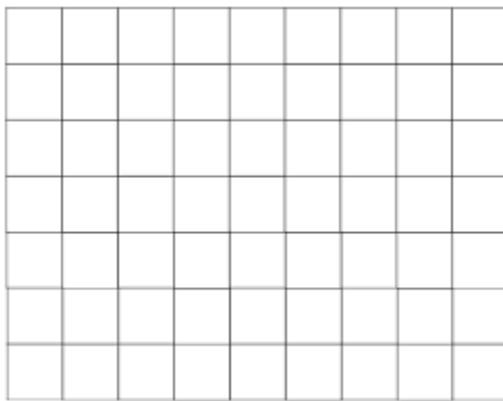


3. Enlargement Transformation

- In the enlargement transformation the object is enlarged, the image is **geometrically similar** to the object (maybe be larger or smaller in size).
- For enlargement transformation, we are given the **position of the centre of enlargement** and the **scale factor, k** .

(I) Enlargement Transformation when the scale factor $k > 0$

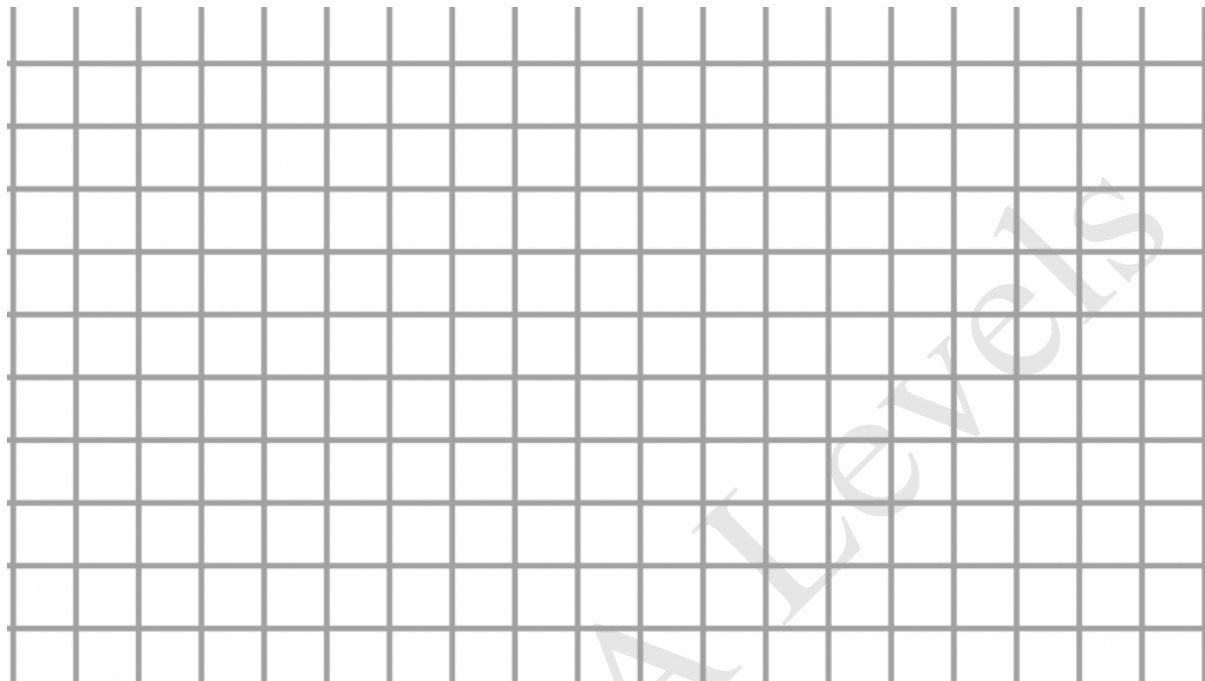
- Given the centre of enlargement origin (0,0) and the scale factor $k > 0$, then the object and image are on the **same** side of the origin.





(II) Enlargement Transformation when the scale factor $k < 0$

- Given the centre of enlargement origin (0,0) and the scale factor $k < 0$, then the object and image are on the **opposite** side of the origin.

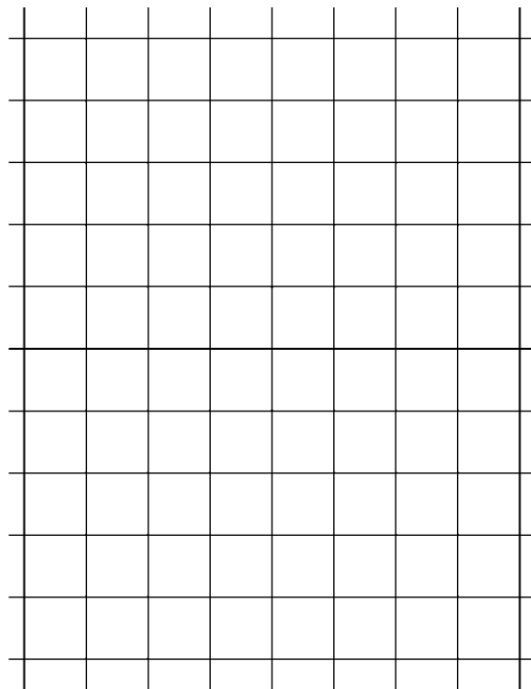


Note:

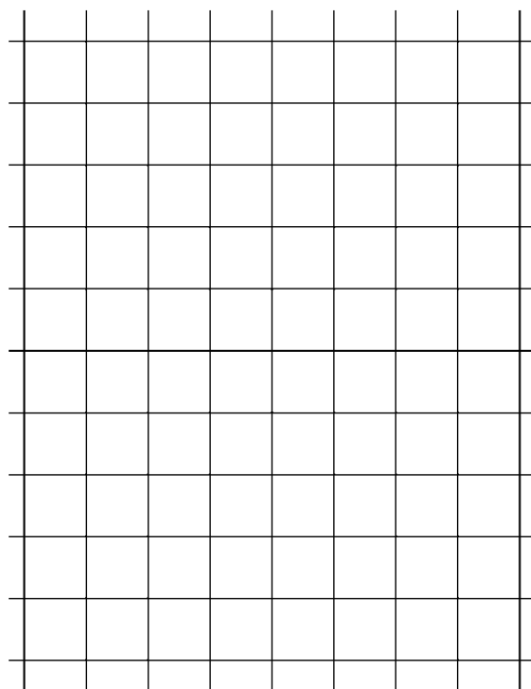


4. Stretching Transformation

(I) Scale factor k parallel to the x -axis



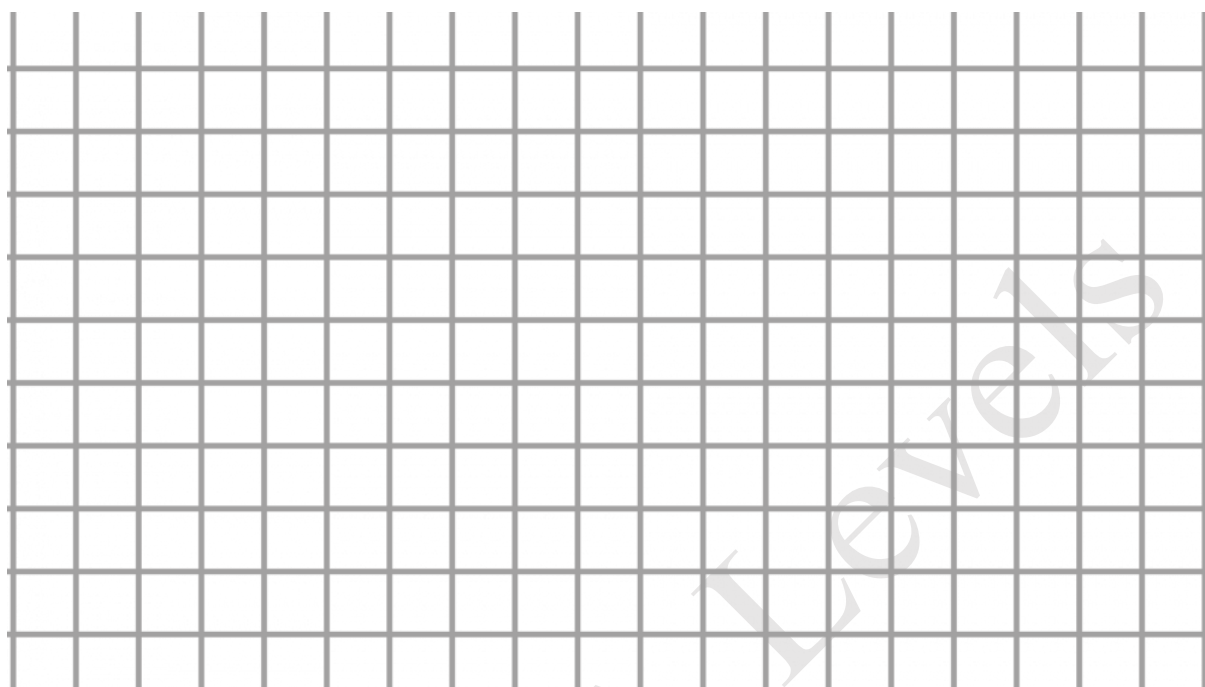
(II) Scale factor k parallel to the y -axis





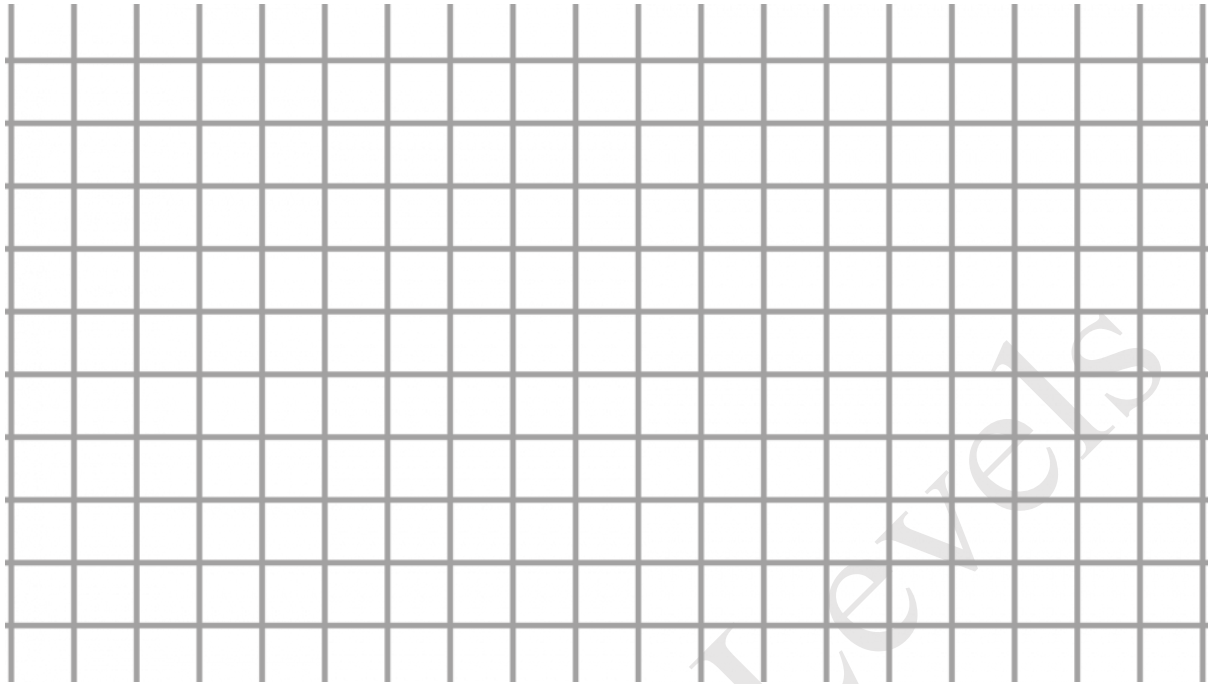
5. Shear Transformation

(I) Shear transformation parallel to the x -axis





(II) Shear transformation parallel to the y -axis



6. Combined Transformations:

- Sometimes question may ask you to find the combined transformation in a matrix.
- For **example 1**, the matrix C represent a stretch by a scale of 2 in the direction of the x -axis, followed by an anticlockwise rotation through 90° about the origin.
- A stretch by a scale factor of 2 in the direction of the x -axis is represented by the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ and an anticlockwise rotation through 90° about the origin is represented by the matrix $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.
- $C = BA = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$.
- Hence the transformation represented by A followed by the transformation represented by B is equivalent to the transformation represented by the matrix BA .
- For **example 2**, let the matrix A represents rotation of the plane anticlockwise through angle of θ about the origin. And let the matrix B represents rotation of the plane anticlockwise through angle of β about the origin. The effect of A followed by B will be the product of BA .

$$BA = \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \cos(\theta + \beta) & -\sin(\theta + \beta) \\ \sin(\theta + \beta) & \cos(\theta + \beta) \end{pmatrix}$$

- For example 3, let the matrix A represents rotation of the plane anticlockwise through angle of θ about the origin. And let the matrix B represents reflection in the line $y = (\tan \beta)x$. The combined transformation would be:

$$BA = \begin{pmatrix} \cos 2\beta & \sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

- **Example 11:**

Reflection in x -axis is given by $M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Find M^2 , what transformation is represented by M^2 ?

- **Example 12:**

The transformation represented by $C = \begin{pmatrix} 0 & 3 \\ -1 & 0 \end{pmatrix}$ is equivalent a single transformation B followed by a single transformation A . Give geometrical description of a pair of possible transformations B and A and state the matrices that represent them.

- **Example 13:**

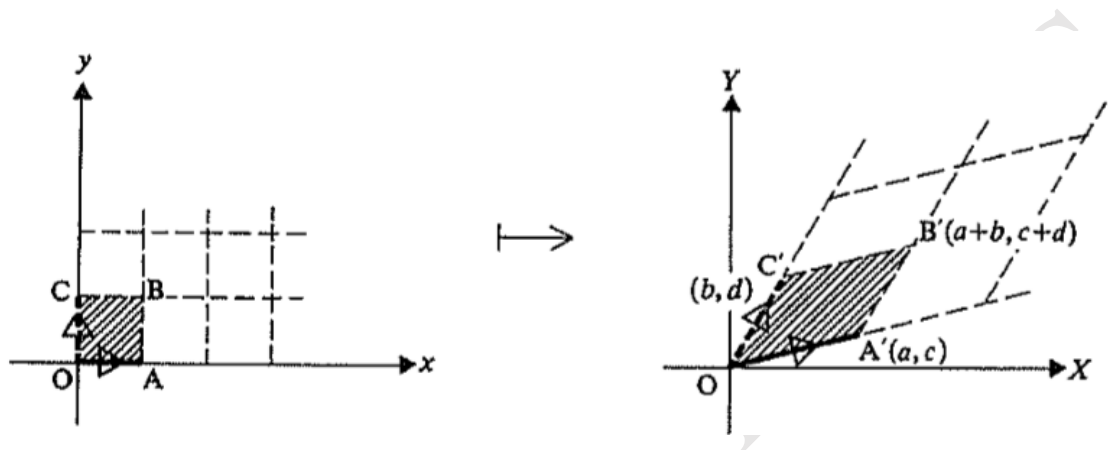
The matrix P represents the rotation of the plane anti-clockwise through an angle of 30° about the origin. The matrix Q represents the reflection in the line $y = x$. Write down 2×2 matrices P and Q .

Find the images of $A(2,3)$ under the transformation given by PQ and QP respectively. Find the matrix R such that $PQ = RQP$.

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7. Relationship between the area scale factor of a transformation and the determinant of the matrix that represents the transformation

- Consider the effect upon the area of a figure produced by the transformation matrix $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.
- $\mathbf{M}$ maps $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to $\begin{pmatrix} b \\ d \end{pmatrix}$.



- From the diagram, the area of $OABC$ is 1 square unit and the area of $OA'B'C'$ is $ad - bc$ square units.
- Hence $\mathbf{M}$ maps an area of 1 square unit in the x - y plane to an area of $(ad - bc)$ square units in the transformed plane. Note that $(ad - bc) = \det(\mathbf{M})$.
- For a transformation represented by a matrix $\mathbf{M}$,

$$\text{Area of image} = \text{Area of object} \times |\det(\mathbf{M})|$$
- If $\det(\mathbf{M})$ is **negative**, the transformation turns an area over.
- If $\det(\mathbf{M}) = 0$ or $\mathbf{M}$ is **singular**, then the transformation maps the x - y plane onto a line or onto the origin.

- **Example 14:**

The triangle OAB is mapped by the matrix $\mathbf{M}$ onto a triangle $OA'B'$ where A and B have coordinates $(0,2)$ and $(3,0)$ respectively and $\mathbf{M} = \begin{pmatrix} 3 & 2 \\ 5 & 8 \end{pmatrix}$. Find the area of the triangle $OA'B'$.

- **Example 15:**

The square whose vertices are the points $A(0,3)$, $B(1,1)$, $C(3,2)$ and $D(3,2)$ is mapped to the plane figure $A'B'C'D'$ by the matrix $\mathbf{M} = \begin{pmatrix} 1 & 2 \\ 1 & -2 \end{pmatrix}$. Find the area of $A'B'C'D'$.



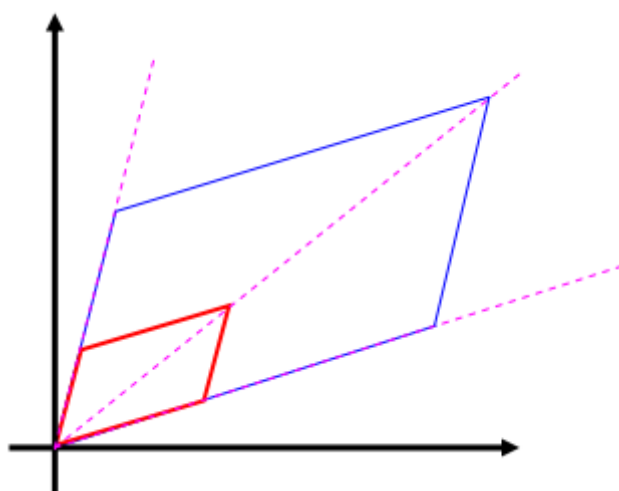
8. Invariant Points:

- The points whose images under a transformation are same as the point (object), are called the **invariant points**.

(i) In **enlargement transformations**, origin is always an invariant point.

For $M \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is a transformation matrix then,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



(ii) In **reflection transformations**: the points on the mirror line are invariant points.

Example,

- (iii) Consider transformation matrix $\begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$, find the invariant point $\begin{pmatrix} x \\ y \end{pmatrix}$.

- **Example 16:**

Find the invariant points under the transformation $\begin{pmatrix} 4 & 1 \\ 6 & 3 \end{pmatrix}$.

- **Example 17:**

Find the invariant points of the transformation represented by the matrix

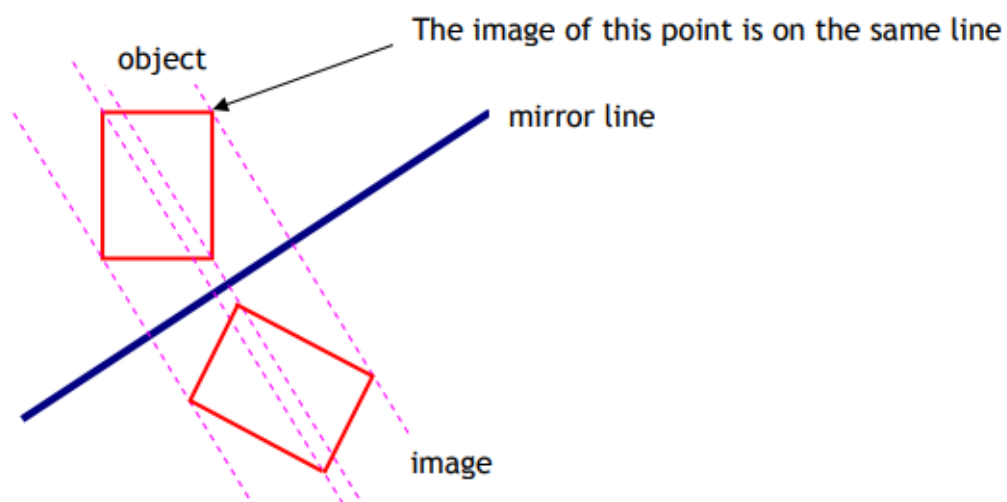
$$\begin{pmatrix} 4 & -1 \\ 2 & 5 \end{pmatrix}.$$

- **Example 18:**

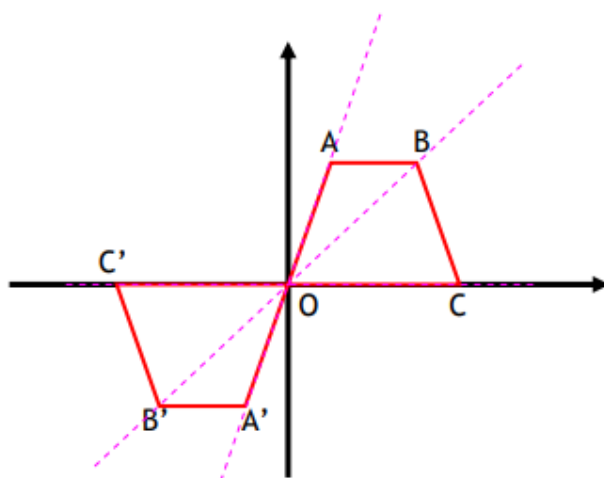
Find the invariant points of the transformation represented by the matrix $\begin{pmatrix} 6 & 5 \\ 2 & 3 \end{pmatrix}$.

9. Invariant Lines:

- A line is said to be invariant for a given transformation if the **image of every point** on the line also **lies on the same line**.
- In **reflection transformation** the image of every point on the **mirror line** remains on the mirror line only, hence the mirror line are **invariant lines**.
- In reflection transformation the lines **perpendicular** to the mirror lines are also invariant lines.



- For most **rotations transformation**, there are no invariant lines **except** a rotation of 180° . In this case, any line that passes through the origin is an invariant line, even though the origin is the only invariant point.



Again, any line through $(0,0)$ is an invariant line.

Just as before, we would say that any line of the form $y = kx$ is an invariant line.



- To find the equations of invariant lines **through the origin** under a transformation:

Let $y = mx$ be the equation of an invariant line under a transformation represented by the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Hence a point (t, mt) on this line.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} t \\ mt \end{pmatrix} = \begin{pmatrix} T \\ mT \end{pmatrix}$$

$$at + mbt = T \quad (1)$$

$$ct + mdt = mT \quad (2)$$

$$(1) \div (2), \frac{a + mb}{c + md} = \frac{1}{m} \Rightarrow bm^2 + (a - d)m - c = 0$$

- (i) If $(a - d)^2 - 4b(-c) < 0$, then the transformation has **no** invariant line.
- (ii) If $(a - d)^2 - 4b(-c) = 0$, then the transformation has **one** invariant line.
- (iii) If $(a - d)^2 - 4b(-c) > 0$, then the transformation has **two** invariant lines.

- **Example 19:**

For matrix $M = \begin{pmatrix} 4 & 11 \\ 11 & 4 \end{pmatrix}$, find the invariant lines, through the origin of the transformation M .

- **Example 20:**

Find the invariant lines of the transformation, through the origin represented by the matrix $\begin{pmatrix} 4 & -1 \\ 2 & 5 \end{pmatrix}$.

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- **Example 21:**

Find the invariant lines of the transformation, through the origin represented by the matrix $\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$.

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Problem Solving:**- Example 22:**

Given $A = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$,

- (i) Describe fully the single geometrical transformation U represented by the matrix A .

The transformation V represent by the 2×2 matrix B , in a reflection in the line $y = -x$.

- (ii) Write down the matrix B .

Given that U followed by V is the transformation T , which is represented by matrix C .

- (iii) Find the matrix C .
- (iv) Show that there is a real number k for which the point $(1, k)$ is invariant under T .

- **Example 23:**

Given $M = \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$,

- (i) Show that M is non-singular.

The hexagon R is transformed to the hexagon S by the transformation represented by the matrix M .

- (ii) Find the area of hexagon S .

The matrix M represents an enlargement with centre $(0,0)$ and scale factor $k > 0$, followed by a rotation anti-clockwise through an angle θ about $(0,0)$.

- (iii) Find the value of k .

- (iv) Find the value of θ .

- **Example 24 (Specimen 2020/1/5):**

The matrix $\mathbf{A}$ is given by $\mathbf{A} = \begin{pmatrix} 5 & k \\ -3 & -4 \end{pmatrix}$.

(a) Find the value of k for which $\mathbf{A}$ is singular. [2]

It is now given $k = 6$ so that $\mathbf{A} = \begin{pmatrix} 5 & 6 \\ -3 & -4 \end{pmatrix}$.

(b) Find the equations of the invariant lines, through the origin, of the transformation in the x - y plane represented by $\mathbf{A}$. [6]

(c) The triangle DEF in the x - y plane is transformed by $\mathbf{A}$ onto triangle PQR .

(i) Given that the area of triangle DEF is 10 cm^2 , find the area of triangle PQR . [2]

(ii) Find the matrix which transform triangle PQR onto triangle DEF . [2]

10. Inverse Transformation

- If a transformation is represented by a matrix $\mathbf{M}$, then $\mathbf{M}$ maps any point (x, y) to a point (x', y') . The inverse transformation represented by $\mathbf{M}^{-1}$ maps the point (x', y') back to the point (x, y) .
- For example, if $\mathbf{M}$ represents an **anticlockwise** rotation through an angle θ about the origin, $\mathbf{M}^{-1}$ represents a **clockwise** rotation through an θ about the origin.
- If $\mathbf{M}$ represents an **enlargement** along the x -axis with scaling factor h , then $\mathbf{M}^{-1}$ represents a **contraction** along the x -axis with a scaling factor $\frac{1}{h}$.
- It should be clear that if the plane is transformed by $\mathbf{M}$, and this operation is immediately followed by $\mathbf{M}^{-1}$, the net effect of the combined transformation is to leave each point unchanged. This fact can be expressed in the following way: $\mathbf{M}^{-1}\mathbf{M} = \mathbf{I}$ or $\mathbf{M}^{-1} = \mathbf{M}$. This transformation is called **self-inverse**.

- **Exercises:**

1. The matrix $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ represents a transformation P .

(i) Describe fully the transformation P .

The matrix $\mathbf{M}$ is given by $\mathbf{M} = \begin{pmatrix} -3 & -1 \\ -1 & 0 \end{pmatrix}$.

(ii) Given that $\mathbf{M}$ represents transformation Q followed by transformation P , find the matrix that represents the transformation Q and describe fully the transformation Q .

2. $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 3 & 6 \\ 11 & -8 \end{pmatrix}$

(i) Find $\mathbf{A}^{-1}$.

The transformation represented by the matrix $\mathbf{B}$ followed by the transformation represented by the matrix $\mathbf{A}$ is equivalent to the transformation represented by the matrix $\mathbf{P}$.

(ii) Find $\mathbf{B}$, giving your answer in its simplest form.

3. Find the matrix that represents a rotation through 90° clockwise about the origin. Find the matrix that represents a reflection in the x -axis.

Hence find the matrix that represents a rotation through 90° clockwise about the origin, followed by a reflection in the x -axis. Hence describe a single transformation that is represented by this matrix.

4. The matrix $\mathbf{A}$ is defined by $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

(i) Describe fully the single transformation represented by the matrix $\mathbf{A}^2$.

(ii) Given that matrix $\mathbf{B}$ represents a reflection in the line $x + \sqrt{3}y = 0$, find the matrix $\mathbf{B}$, giving the exact values of any trigonometric expressions.

(iii) Hence find the coordinates of the point P which is mapped onto $(0, -4)$ under the transformation represented by $\mathbf{A}^2$ followed by a reflection in the line $x + \sqrt{3}y = 0$.

5. The matrix $\mathbf{A}$ is defined by $\mathbf{A} = \begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}$.

(i) Calculate the matrix $\mathbf{A}^2$.

(ii) Show that $\mathbf{A}^3 = k\mathbf{I}$, where k is an integer.

(iii) Describe the single transformation, or combination of two transformations, corresponding to each of the matrices: (a) $\mathbf{A}^3$ (b) $\mathbf{A}$.

6. Let $\mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$.

(i) Describe fully the single transformation U represented by $\mathbf{P}$.

(ii) The transformation V represented by the matrix $\mathbf{Q}$ is a reflection in the line with the equation $y = x$.

(iii) Write down the matrix $\mathbf{Q}$.

(iv) The transformation U followed by the transformation V is the transformation T represented by the matrix $\mathbf{R}$. Find $\mathbf{R}$ and deduce that the transformation T is self-inverse.

7. A transformation in the $x - y$ plane is represented by the matrix $\mathbf{A} = \begin{pmatrix} 3 & a \\ 6 & 2a \end{pmatrix}$.

Show that

(i) all points in the plane are mapped by $\mathbf{A}$ to points of a certain line, and state the equation of the line.

(ii) all points which are mapped by $\mathbf{A}$ to the origin lie on certain line, and state the equation of the line.



8. If a and b are non-zero, the matrix $\mathbf{A} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ represents a shear with x -axis invariant, and the matrix $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$ represents a shear with y -axis invariant. Show that the application of these shears is not commutative.

Find the two invariant lines, in the form $y = mx$, of the transformation represented by $\mathbf{BA}$ when $a = 4, b = 2$.

9. Let $\mathbf{A} = \begin{pmatrix} 5k & 3k - 1 \\ -3 & k + 1 \end{pmatrix}$.

(i) Given that $\mathbf{A}$ is singular, find the possible value of k .

It is now given that $k = 2$.

(ii) Show that there are no invariant lines, through the origin, of the transformation represented by $\mathbf{A}$.

A triangle T is transformed onto a triangle T' by the transformation represented by $\mathbf{A}$. The vertices of T' have coordinates $(0,0)$, $(-20,6)$ and $(10c,6c)$, where c is a positive constant.

The area of the triangle T' is 135 square units.

(iii) Find the coordinates of the vertices of the triangle T , in terms of c where necessary and hence find the value of c .

SUMMARY: TABLE OF TRANSFORMATIONS

Transformation	Object	Image	Matrix
1. REFLECTION			
Reflection in x -axis	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} x \\ -y \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection in y -axis	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} -x \\ y \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
Reflection in origin	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} -x \\ -y \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection in the line $y = x$	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} y \\ x \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Reflection in the line $y = -x$	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} -y \\ -x \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$
Reflection in the line $y = x \tan \theta$	$A \begin{pmatrix} x \\ y \end{pmatrix}$	A'	$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$
Reflection in the line $y = mx$	$A \begin{pmatrix} x \\ y \end{pmatrix}$	A'	$\begin{pmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & \frac{1-m^2}{1+m^2} \end{pmatrix}$
2. ROTATION			
Through θ anticlockwise at origin	$A \begin{pmatrix} x \\ y \end{pmatrix}$	A'	$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
Through 90° anticlockwise at origin	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} -y \\ x \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$
Through 90° clockwise at origin	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} y \\ -x \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
Through θ clockwise at origin	$A \begin{pmatrix} x \\ y \end{pmatrix}$	A'	$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$
3. ENLARGEMENT			
Centre of enlargement at origin, scale factor $k > 0$	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} kx \\ ky \end{pmatrix}$	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$



Centre of enlargement at origin, scale factor $k < 0$	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} kx \\ ky \end{pmatrix}$	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$
4. STRECHING			
Scale factor k , parallel to x -axis	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} kx \\ y \end{pmatrix}$	$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$
Scale factor k , parallel to y -axis	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} x \\ ky \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$
5. SHEARING			
Parallel to x -axis	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} x + ky \\ y \end{pmatrix}$	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$
Parallel to y -axis	$A \begin{pmatrix} x \\ y \end{pmatrix}$	$A' \begin{pmatrix} x \\ kx + y \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$
6. Area of Image = M x Area of object, where M is the determinant of matrix.			
7. Invariant points: The point for which the image is same as the object for a given transformation. $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$			
8. Invariant lines for a given transformation: If the image of every point on this line lies on the same line for every point on line $y = mx + c$, (x', y') lies on $y = mx + c$. i.e. $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$			